Monday Warm Up: Unit 4, Magnetic Induction

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1 Memory Bank

- $\epsilon = -N\Delta\phi_m/\Delta t$ ... Faraday's Law
- $\phi_m = \vec{B}\cdot\vec{A}$ ... Definition of magnetic flux
- 1 T = 1 N A^{-1} m^{-1} ... Definition of 1 Tesla
- $\epsilon = NAB\omega$ ... Peak emf of electric generator

2 Faraday's Law

1. Suppose a uniform B-field of 0.1 T passes through a wire loop with an area of 0.05 m^2. Calculate ϕ_m, the magnetic flux, if the angle between the B-field and loop area is (a) 0 degrees, (b) 45 degrees, and (c) 90 degrees. (d) Suppose the B-field drops to 0 T in 1 ms. What is the induced emf?

2. Verify that the units of $\Delta\phi_m/\Delta t$ are volts.

3. A coil is moved through a magnetic field as shown in Fig. 1. The field is uniform inside the rectangle and zero outside. What is the direction of the induced current and what is the direction of the magnetic force on the coil at each position shown?

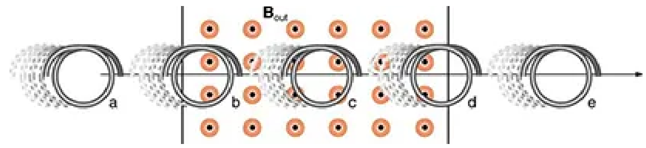

Figure 1: A loop passes through a B-field.

3 Electric Generators

1. (a) A bicycle generator rotates at 1875 rad/s, producing an 18.0 V peak emf. It has a 1.00 by 3.00 cm rectangular coil in a 0.640 T field. How many turns are in the coil? (b) Is this number of turns of wire practical for a 1.00 by 3.00 cm coil?